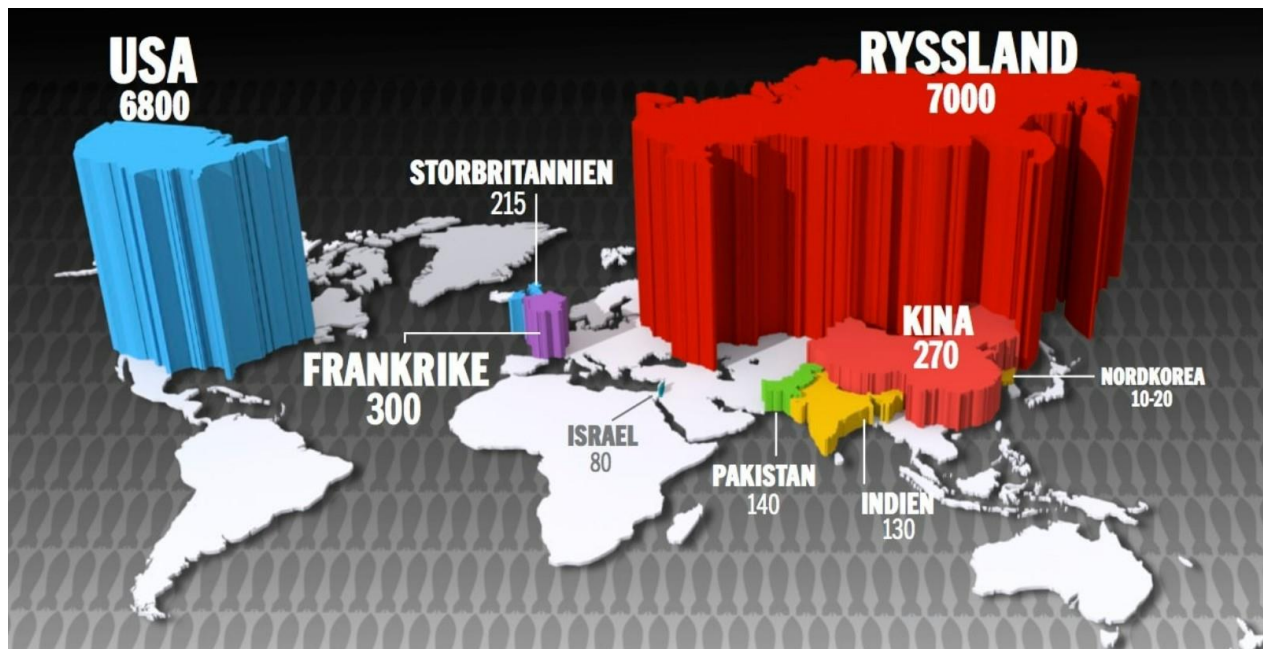


Assignment 9

1. Bad Data Visualization Example

The visualization below is a 3D map showing nuclear weapons by country, originally created based on SIPRI (Stockholm International Peace Research Institute) data.



2. Problems in This Visualization

Problem 1: Violation of the Proportional Ink Principle

In this map, Russia has 7,000 nuclear weapons and the USA has 6,800- a difference of only 3%. However, Russia appears approximately 3 times larger than the USA due to its geographical size being used as the base of the 3D columns.

Problem 2: Misleading 3D Representation

3D visualizations create perspective distortion where distant objects appear smaller. This makes accurate comparison between values nearly impossible. The depth perception interferes with the viewer's ability to accurately assess quantities.

Problem 3: Geographic Distortion

Using country borders as the base of 3D columns conflates geographic area with data values. Russia's geographic area is about 1.7 times larger than the USA, causing its visual representation to be disproportionately inflated regardless of the actual data.

Problem 4: Unnecessary Complexity

A world map and 3D effects are not needed to present this data effectively. The added visual complexity obscures the main message rather than clarifying it.

3. Solution:

The cleanest solution is a simple bar chart. When the Russia and US bar charts are placed side-by-side, it's clearly visible how small the difference between 7000 and 6800 is.

Values can be added to the ends of the bars for easier reading.

2D design will eliminate perspective distortion.

References

<https://www.aftonbladet.se/kultur/a/4dJE4g/karnvapenhotet-har-aldrig-varit-storre>

